

Part 1 AI Reflection

The feature I chose is moving all game state to the server and adding a turn based computer opponent. I removed client side state and made the server responsible for ship placement, hits, turn order, and win conditions. I also added New Game and Restart Same Board with clear reset rules. Then I added a computer opponent that fires back after the player. The server enforces turn order and runs simple hunt logic.

The feature makes the server fully in control. The client only renders the board, sends shot coordinates, and shows results. The server checks moves, updates the boards, controls turns, runs the AI, and decides when the game ends. New Game resets everything. Restart Same Board keeps the ships but clears hits and resets turns.

I chose this because earlier versions had logic in JavaScript and caused desync and reset bugs. Moving all state to the server fixed that. Adding the AI forced me to think about game flow and rule enforcement instead of just UI behavior.

All real state lives in the PHP session. That includes both boards, hits, turn state, AI memory, and winner status. The client does not control any game rules.

I used AI to help structure the system. First I asked it to generate a version where the server owns all state. When state was disappearing, I asked it to explain the flow. That helped me see what was wrong. Later I asked it to define what should reset and what should not. Then I asked it to add a turn based computer opponent enforced on the server.

My prompts changed over time. At first I asked for structure. Later I asked about state rules and boundaries. I stopped asking AI to just add features and started asking how the system should behave.

Manually, I refactored the session variables, removed client side rule checks, added a turn flag, blocked invalid shots, and made sure AI memory persisted between turns. I reviewed and adjusted all AI output to match my architecture.

Prompt Log

Prompt 1

Build Battleship so the server owns all state and the client only renders and sends actions.

Why: I wanted correct structure from the start.

Change: Used as a base.

Prompt 2

Why is my state changing or disappearing after each shot.

Why: The game was inconsistent.

Change: Centralized state updates on the server.

Prompt 3

Define what changes for New Game and Restart Same Board.

Why: Needed clear reset rules.

Change: Implemented separate reset logic on the server.

Prompt 4

Add a turn based computer opponent enforced on the server.

Why: Needed proper turn control.

Change: Added turn flag and server side AI memory.

Prompt 5

Simplify the CSS.

Why: Focus on gameplay not visuals.